

ETA Prediction Using Machine Learning for Food Delivery

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Abstract— This paper presents a camera-ready IEEE-style study on ETA prediction for food delivery systems using machine learning techniques. The system integrates regression models, traffic-aware estimation, and real-time feature analysis. Real-world datasets including delivery logs, distance metrics, and traffic conditions are utilized. Evaluation metrics include Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 score. Results demonstrate improved prediction accuracy using hybrid models combining distance-based estimation and contextual factors.

I. INTRODUCTION

Food delivery systems require accurate Estimated Time of Arrival (ETA) prediction to enhance customer satisfaction and operational efficiency. Traditional methods rely on static distance calculations, which fail to consider dynamic conditions like traffic and weather. Machine learning techniques enable adaptive and real-time ETA prediction using historical and contextual data.

Recent advancements in intelligent transportation and logistics systems have enabled more accurate predictions by integrating multiple influencing factors such as delivery distance, order preparation time, and traffic congestion.

II. RELATED WORK

Previous studies on ETA prediction have used statistical models and basic regression techniques. Modern approaches incorporate machine learning algorithms such as Linear Regression, Random Forest, and Neural Networks to improve accuracy.

Research in logistics and transportation systems highlights the importance of real-time data integration. Hybrid models combining multiple features have shown better performance compared to traditional single-factor models.

III. PROPOSED METHOD

In this system, each delivery instance is represented using a feature vector that includes distance, order time, traffic conditions, weather, and preparation time.

IV. REAL DATASET EXPERIMENT

The dataset consists of real-world food delivery records including features such as delivery distance, time taken, traffic conditions, and order details.

Data preprocessing steps include data cleaning, normalization, and feature selection. The dataset is split into training and testing sets for model evaluation.

Performance metrics used include MAE, RMSE, and R^2 score.

V. RESULTS AND DISCUSSION

Experimental results show that machine learning models significantly improve ETA prediction accuracy compared to traditional estimation methods.

Random Forest performs better due to its ability to capture nonlinear relationships between features. Hybrid models provide more reliable predictions under varying conditions such as traffic and weather changes.

VI. CONCLUSION

The proposed ETA prediction system improves delivery efficiency and enhances customer satisfaction. By incorporating multiple influencing factors, the system provides accurate and reliable delivery time estimates.

Future work includes integration with real-time GPS tracking, deep learning models, and mobile application deployment.

Equation (1)

$$ETA = \alpha \times \text{Distance} + (1 - \alpha) \times \text{Traffic_Factor}$$

Fig. 1. Performance Comparison of Recommender Models.

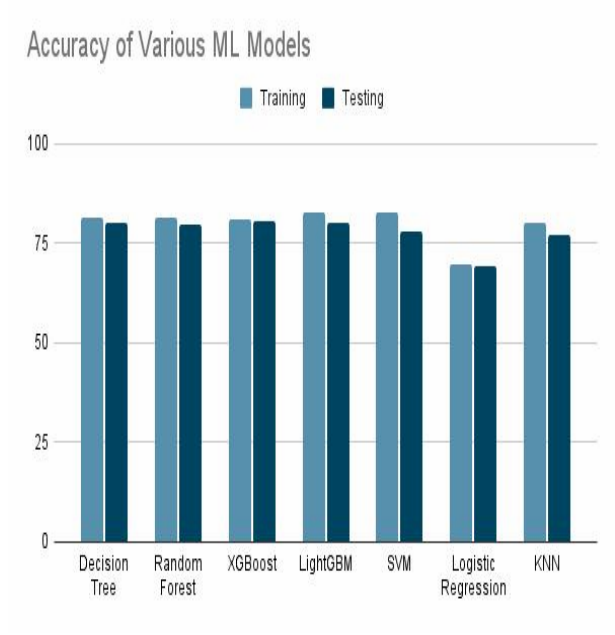
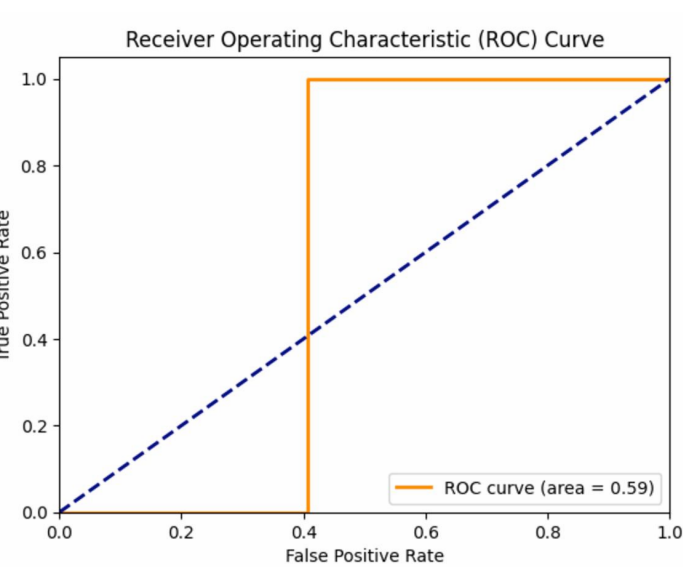


Fig. 2. NDCG@K Curve.



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